Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/07/2017
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	GCR Simulator Validation Studies with Human and Mouse Models		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	02/04/2016	**End Date:**	02/03/2019
No. of Post Docs:		**No. of PhD Degrees:**	4
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	No key personnel changes.		
COI Name (Institution):	Turker, Mitchell (Oregon Health & Science University) Grossi, Gianfranco (Universita Federico II) Guida, Peter (Brookhaven National Laboratory) Grygoryev, Dmytro (Oregon Health and Science University) Rwatambuga, Furaha (Oregon Health and Science University) Bonti, Kristi (Brookhaven National Laboratory) Gauny, Stacey (Lawrence Berkeley National Laboratory)		
Grant/Contract No.:	NNJ16HP23I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project addresses NASA's request to help design and validate its planned galactic cosmic radiation (GCR) simulator. Space radiation has two major components: GCR, which is comprised of highly energetic ions from protons through uranium, and solar particle events, which are primarily moderate energy protons. Most currently funded NASA ground-based studies in radiobiology examine one highly energetic ion at a time because of facility limitations. While this approach provides insight into the biological effect of each particle, insight into potential interactions from particles of different energies or charges is missed and the deep space environment is not accurately copied. Recent improvements in facility capability allow mixed beam exposures to more closely approximate the GCR, but questions remain about the final composition of the beams to be included. We will build upon our prior work on single particle-induced mutations in cultured cells and mice (cancer-relevant endpoints). Work with cultured cells will be performed because it will provide data more quickly, aid with cross-species extrapolation of risks, and inform the mouse work. We will first measure the effects of acute exposures to alpha particles (charged helium nuclei) because basic information about this GCR component is lacking. Next, we will measure the effect of acute exposures to a mixture of protons, alpha particles, and silicon ions to mimic the effect of a simple GCR simulator model, followed by acute exposures to the final GCR mixture that is chosen (informed in part from our first two experiments). Finally, we will determine the effects of a protracted exposure to the GCR simulator, which will reflect more closely exposures that will occur in the space environment. Our results will help NASA to both design and validate the GCR simulator.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The purpose of this project is to provide input to NASA's risk model for cancer regarding exposure to accelerated charged particle radiations that simulate those found in space. The work is designed to address cancer development processes at relatively low doses and fluences that are space-relevant. The project specifically uses different types of charged particle radiations alone or in combination to determine whether it is possible to predict the effects of multiple ions from the effects measured in studies using single light or heavy charged particle beams. This work is of relevance to naturally occurring Earth-based exposures to densely ionizing radiations such as radon. It will also inform our understanding of the mechanisms underlying potential risks to normal tissues associated with the implementation of charged particle radiation therapy.
Task Progress:	This is the first year of this newly funded program to address NASA's needs regarding the design and implementation of a complex set of highly energetic charged particle beams to simulate the galactic cosmic ray environment. In response to the complex radiation field expected in actual flight conditions in deep space and on any deep space bodies where exploration may occur, the first set of studies focused on understanding the genotoxic effects of accelerated helium ions which are among the more abundant ion species in these settings. Studies with murine epithelial cells exposed in culture indicate that helium ions (250 MeV/amu, LET (linear energy transfer) = 1.6 keV/µm) induce dose-dependent cytotoxicity and the induction of mutations at a model autosomal locus, and that helium ions appear to be more effective per unit dose than what we reported previously with the same model system using 1 GeV protons (LET = 0.24 keV/µm). Other work initiated in the first series of runs is still in progress. Additional studies initiated in Fall 2016 used a combination of ion beams administered in rapid succession to measure these genotoxic endpoints. A goal of this study is to test the hypothesis that the genotoxic effects of a simulated GCR exposure can be predicted from the results obtained for individual ions. This will help aid the further development of NASA's risk model for carcinogenesis.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Kronenberg A, Turker M. "GCR simulator validation studies with human and mouse models." Presented at the 27th Annual Space Radiation Investigators' Workshop, Galveston, TX, February 8-11, 2016. 27th Annual Space Radiation Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
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Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Bonti K, Guida P, Grygoryev D, Turker M. "Effect of dose-fractionation on the genotoxicity of energetic Si ions in murine epithelial cells." Presented at the 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. , Oct-2016
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